



Augmented Reality Based on Cockpit Inspection

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List Of Contents

- Problem Statement
- Project Overview
- Augmented Reality
- Required Software's
- Project Process Steps
- Future Scope
- Conclusion

Problem Statement

Traditional cockpit inspection methods are time-consuming and depend heavily on manual checks, increasing the chance of human errors. Complex cockpit controls and long inspection procedures can lead to missed steps, delays, and safety concerns. An efficient and intelligent inspection system is needed to improve accuracy, reduce inspection time, and enhance operational safety.

Project Overview

The AR Inspection System is a smart wearable solution that displays real-time inspection instructions directly in the user's view, making cockpit inspection easier and faster. It helps reduce human errors, save inspection time, and improve safety through a digital inspection process.

Augmented Reality

Augmented Reality (AR) is a technology that overlays digital information, such as 3D models, text, images, or videos, onto the real-world environment in real time. It enhances user interaction by combining physical objects with virtual content, enabling applications in education, healthcare, gaming, maintenance, and aircraft inspection systems.



Augmented Reality

Required Software's

In the software components support image processing, AR visualization, damage detection, and maintenance analysis.

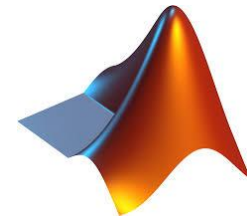
UNITY



Vuforia Engine



MATLAB



Project Process Steps

Unity-Step Process

Unity 3D is a cross-platform software used for developing 3D applications, games, and Augmented Reality (AR) projects. It helps create interactive real-time content for Android, iOS, Windows, and other platforms.

Step 1 – Download Unity Hub:

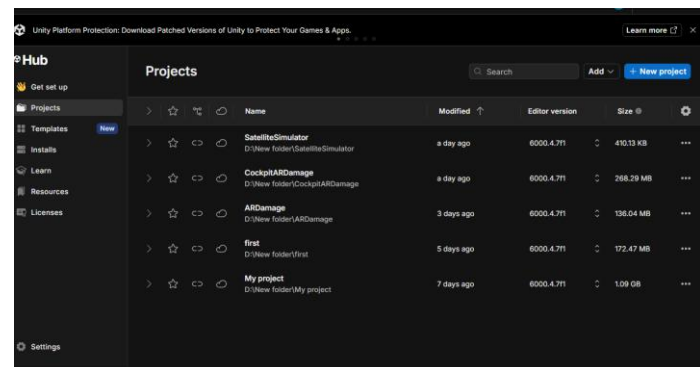
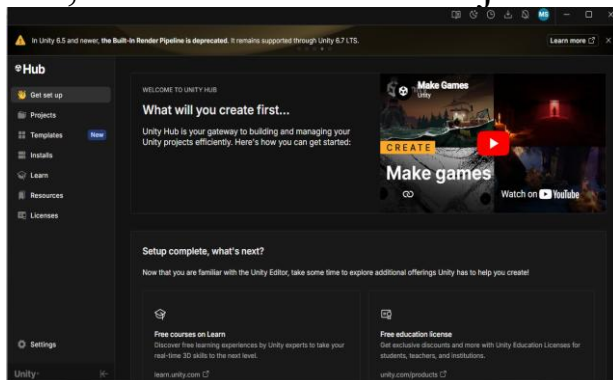
Go to Unity website, create a Unity ID, and download Unity Hub for your operating system.

Step 2 – Install Unity Version:

Open Unity Hub, select a Unity Editor version (example: Unity 2019.4.32f1), and install required modules.

Step 3 – Create New Project:

Open Unity Hub, click New Project, choose 3D Template, enter project name, and click Create Project to start development.



Shows the user interface in Unity Hub.

Vuforia SDK – Steps Process

Vuforia SDK is an Augmented Reality (AR) software development kit used to create AR applications for mobile devices. It uses computer vision and image processing to detect images and 3D objects in real time.

Step 1 – Create Vuforia Account:

Open the Vuforia Developer Portal, create an account, and generate a free license key.

Step 2 – Create Target Database:

Go to **Target Manager**, create a database, and upload target images (.jpg or .png) for AR tracking.

Step 3 – Download and Use Database:

Download the target database and import it into Unity to display 3D models on detected images.

The screenshot shows the 'Basic Plan' page in the Vuforia Developer Portal. It includes a navigation bar with links to Home, News, Downloads, Library, Support, Pricing, My Account, and Log Out. Below the navigation bar, there are tabs for Account, Plan & Licenses, Credentials, and Target Manager. The 'Basic Plan' section describes the features of the Basic Plan and provides a 'Generate Basic License' button. To the right, there is an 'Upgrade to Premium' section with a 'Request Now' button. Below these sections, there is a table listing various target databases.

Name	Type	Status	Date Modified
8945	Basic	Active	May 25, 2026
Altrospection	Basic	Active	May 21, 2026
first	Basic	Active	May 19, 2026
cockpit 777	Basic	Active	May 19, 2026
my	Basic	Active	May 19, 2026
CockpitViewer	Basic	Active	May 11, 2026
Engine	Basic	Active	May 11, 2026
Ar	Basic	Active	May 09, 2026
Do228 AR Maintenance	Basic	Active	May 08, 2026

The screenshot shows the 'Target Manager' page in the Vuforia Developer Portal. It includes a navigation bar with links to Home, News, Downloads, Library, Support, Pricing, My Account, and Log Out. Below the navigation bar, there are tabs for Account, Plan & Licenses, Credentials, and Target Manager. The 'Target Manager' section includes a 'Generate Database' button and a search bar. Below the search bar, there is a table listing various target databases.

Database	Type	Targets	Date Modified
AircraftDB	Device	2	May 21, 2026
BOEING777	Device	1	May 19, 2026
boeing787	Device	1	May 20, 2026
cockpit	Device	1	May 11, 2026
cockpitdatabase	Device	1	May 14, 2026
CockpitDB	Device	4	May 18, 2026
Do228DB	Device	1	May 09, 2026
Do228_Database	Device	1	May 08, 2026
first	Device	1	May 19, 2026
my	Device	2	May 19, 2026

TurboSquid – Simple Steps Process

TurboSquid is an online platform used to download 3D models for games, AR, animation, and simulation projects.

Step 1 – Create Account:

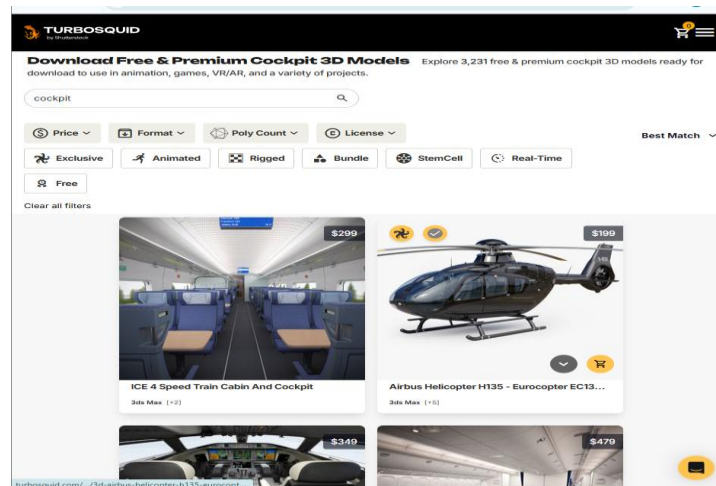
Open TurboSquid website and create an account.

Step 2 – Search 3D Models:

Search required models such as aircraft or helicopters and select the preferred format (.obj, .fbx, etc.).

Step 3 – Download and Import:

Download the model and import it into Unity or Blender for development.



Blender Software – Steps Process

Blender is a free and open-source software used for 3D modeling, animation, rendering, and simulation.

Step 1 – Install Blender:

Download and install Blender software on Windows, macOS, or Linux.

Step 2 – Create or Edit 3D Models:

Use modeling tools to design, modify, or optimize 3D objects.

Step 3 – Export for Unity:

Export the completed 3D model and import it into Unity for AR application development.



Unity + Vuforia +MATLAB AR COCKPIT Project – Steps Process

Step 1 – Create Scene and Add AR Camera

Create a new Unity scene and rename it as “**cockpit view**”. Install the Vuforia Engine package and add an **AR Camera** to the Hierarchy.

Step 2 – Configure Vuforia License and Target Image

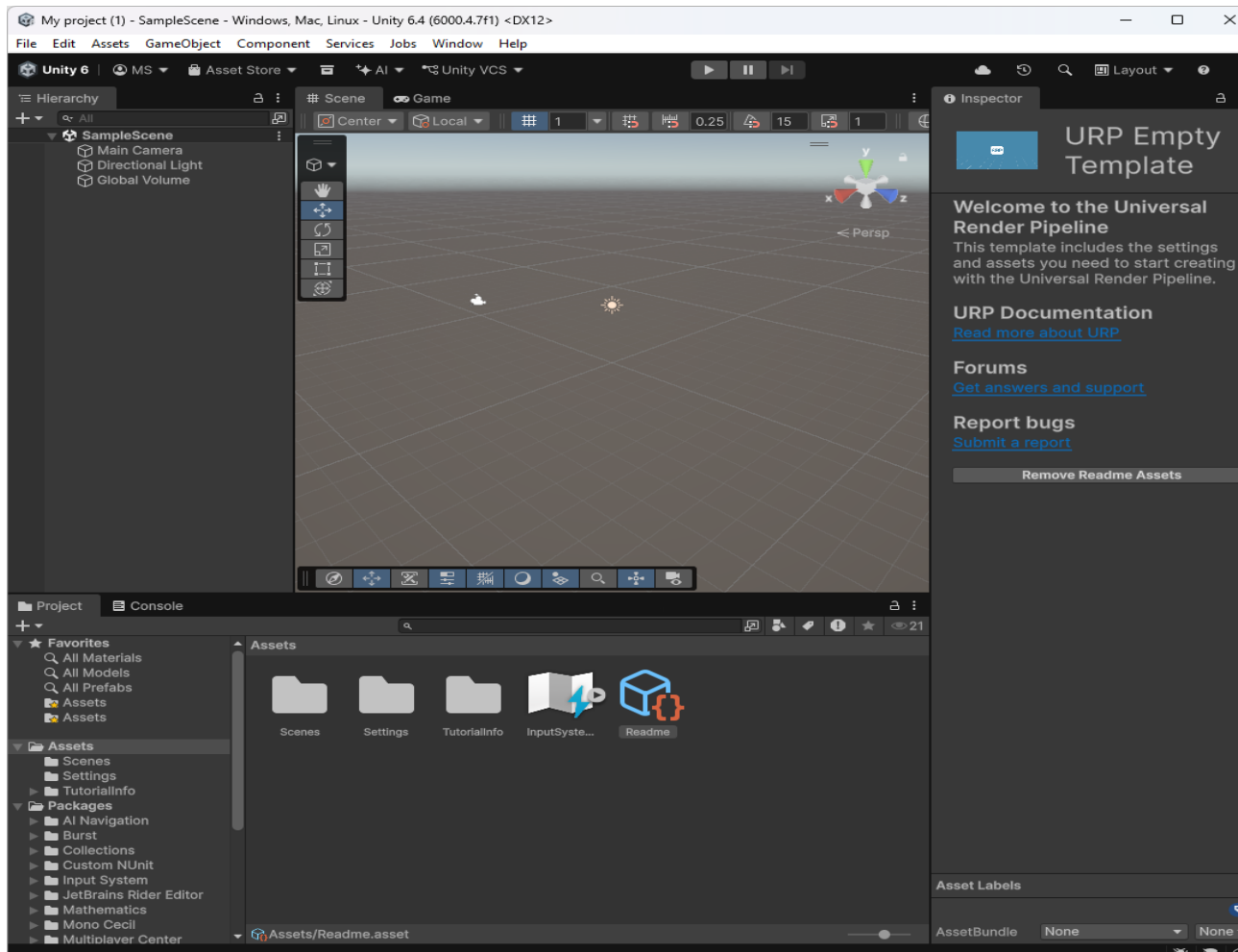
Create a Vuforia account, generate a **License Key**, and add it to the AR Camera configuration. Upload the target image to the Vuforia database and import the database into Unity.

Step 3 – Add 3D Model and Image Target

Import the cockpit 3D model and target database into Assets. Create an **Image Target**, select the database image, and place the 3D model as a child object.

Step4-Create Canvas and UI Controls

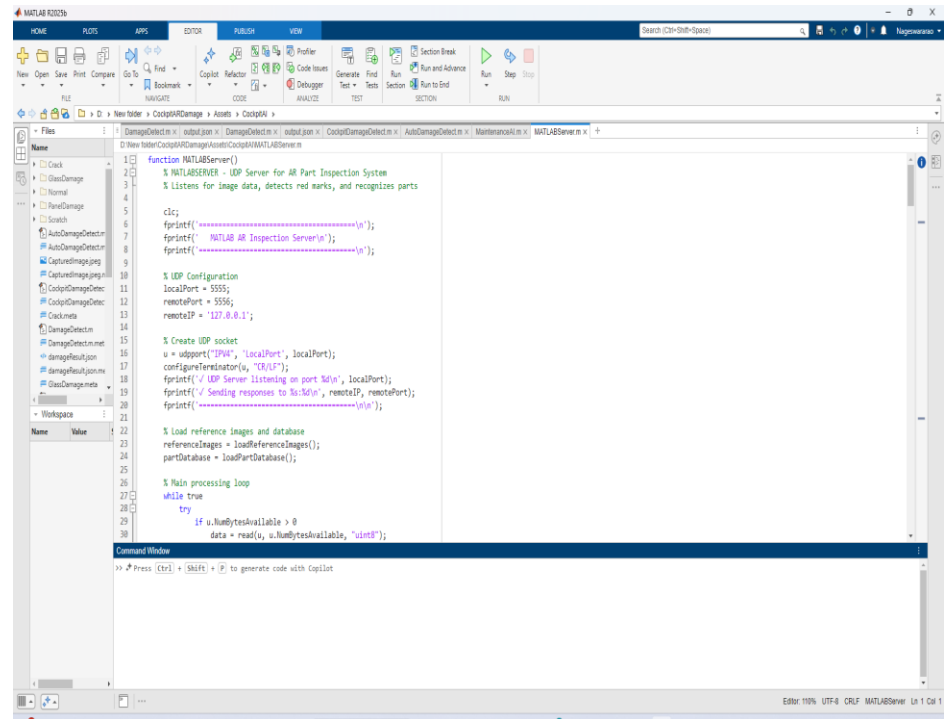
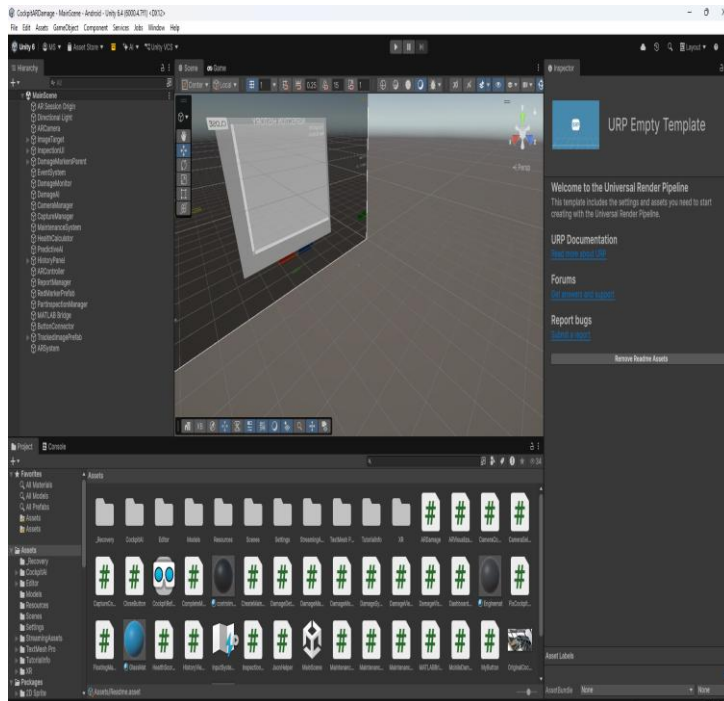
Create a **Canvas** and set the reference resolution to **1920 × 1080**. Add buttons like **Rotate**, **Zoom In**, **Zoom Out**, and **Info** to control the AR aircraft model.

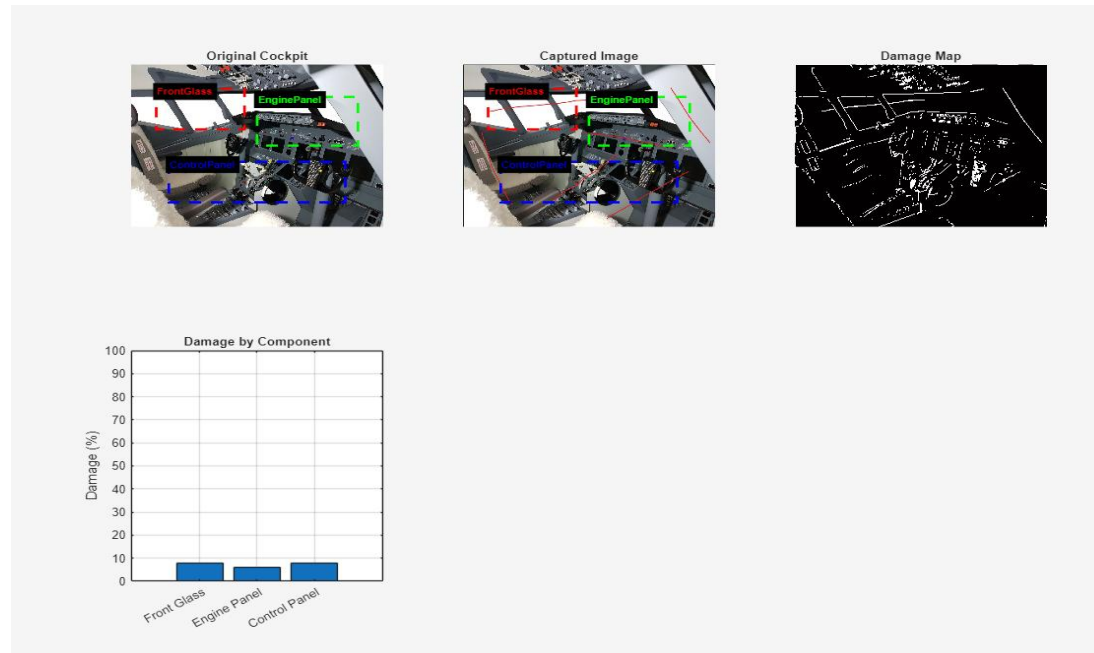


Shows the user interface in Unity Hub.

Step 5:C# Scripts in Unity and MATLAB Image Processing Testing

Created C# scripts in Unity to capture images, connect MATLAB processing, detect damage areas, and display AR indicators on aircraft components. MATLAB image processing was tested by comparing reference and captured images to calculate damage percentage and verify AR visualization results.



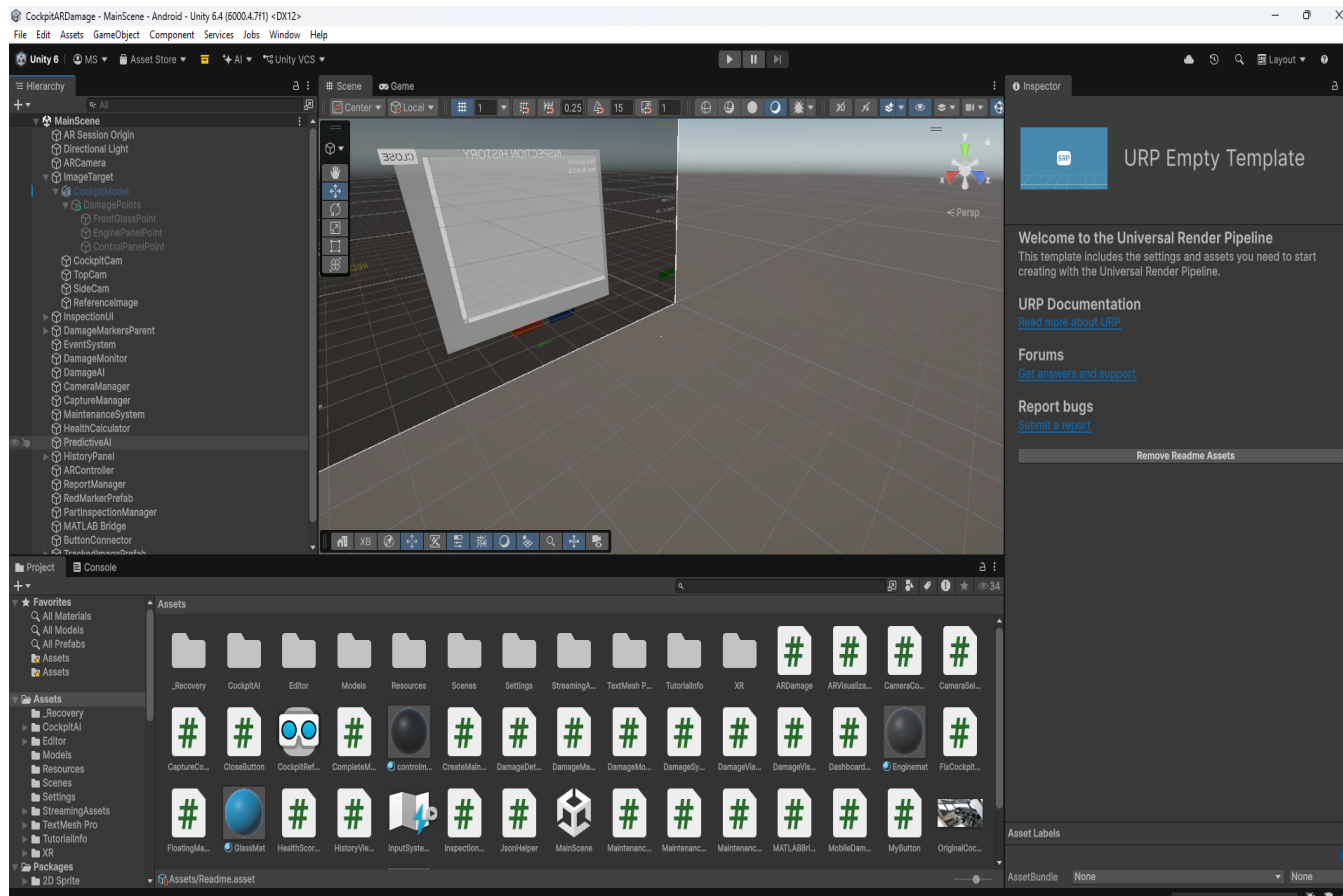


Step6:JSON Output Connection with Unity

After MATLAB image processing completes, the generated output.json file is loaded into Unity using a JSON Loader C# script to transfer damage analysis results. Unity reads the JSON data and updates AR visualization, damage indicators, and inspection information in real time.

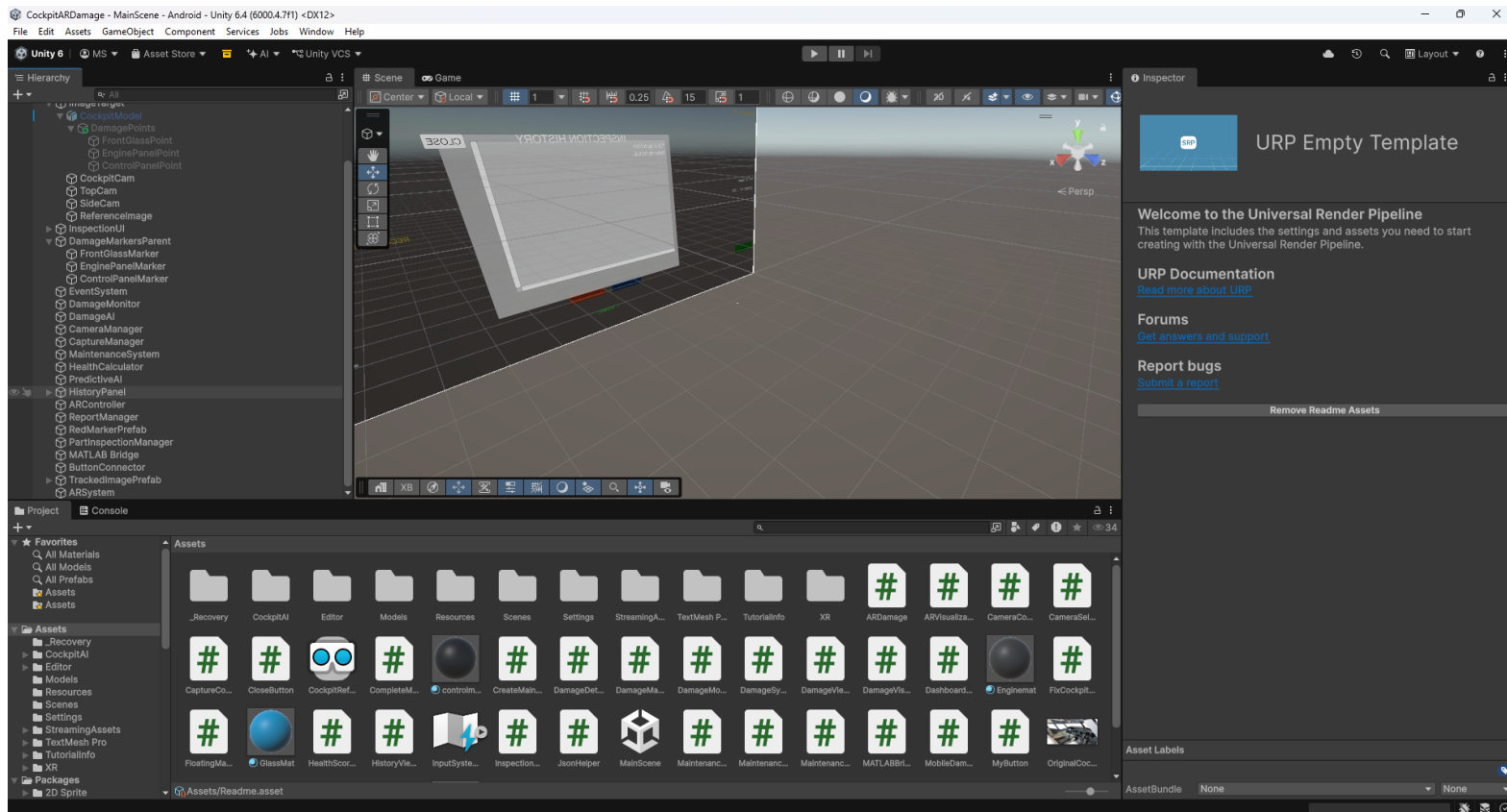
Step7:Material Creation and Damage Point Connection

Created Red Material for damaged areas and Grey Material for normal aircraft components in Unity. Connected damage points such as Front Panel, Glass Panel, and Control Panel as child objects under ImageTarget to enable AR damage visualization and material changes during inspection.



Step8: Multi-Camera Setup and Cockpit Model Connection

Created three cameras (Top Camera, Side Camera, and Front Camera) in Unity to capture cockpit inspection images from multiple viewpoints. Connected all cameras as child objects of the Cockpit Model so camera positions move together with the AR cockpit model during tracking and inspection visualization.



Step 9: Inspection UI and System Components Setup

Created Inspection UI with TopPanel, MiddlePanel, BottomPanel, StatusText, and StatusIndicator to display aircraft inspection information and maintenance status. Added Main Dashboard, HistoryPanel, and DamageMarkersParent for organizing inspection reports and AR damage visualization.

Integrated FrontGlassMarker, EnginePanelMarker, and ControlPanelMarker with RedMarkerPrefab to indicate detected damage locations. Connected system scripts including DamageMonitor, DamageAI, CameraManager, CaptureManager, MaintenanceSystem, HealthCalculator, PredictiveAI, ARController, ReportManager, and PartInspectionManager to support image capture, AI damage analysis, predictive maintenance, and AR cockpit inspection workflow

Step 10: Button connector and AR System Integration

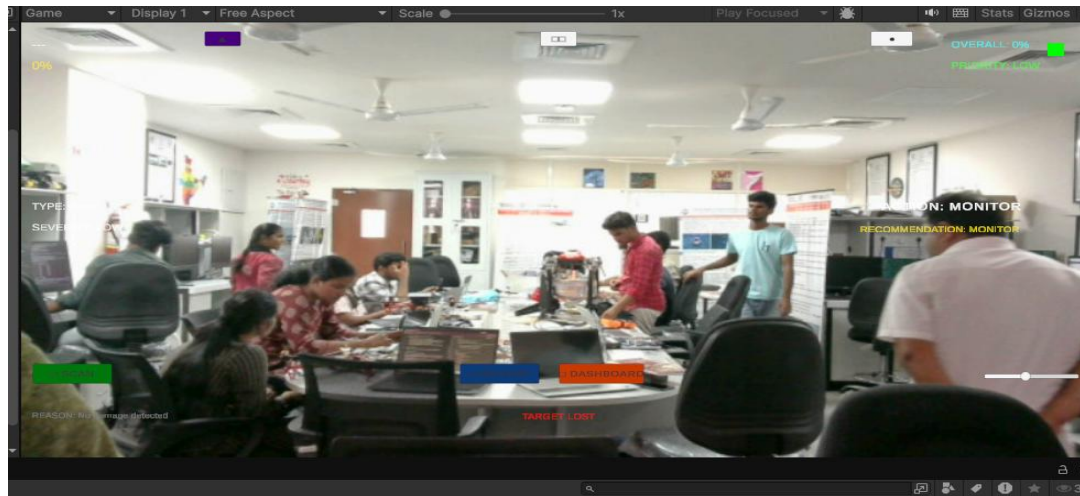
Added ButtonConnector to connect UI buttons with inspection functions, and created TrackedImagePrefab to manage AR image recognition and cockpit model tracking.

Integrated the ARSystem to coordinate image tracking, multi-camera inspection views, MATLAB output processing, and real-time AR visualization for aircraft maintenance and damage inspection workflow.

Step 11 :Final Setup and Testing

Adjust panel positions, add icons and colors, then run the project. When the camera detects the target image, the Cockpit 3D AR model will appear on the screen.





Step12:APK Build Process in Unity

After successful testing, configure Build Settings in Unity by selecting Android Platform, adding scenes, and switching the platform to Android. Open Player Settings, set package name, minimum API level, and enable required permissions for camera and AR functions.

Connect an Android device or select Build APK, choose the output folder, and generate the APK file. Install the APK on the mobile device and perform final testing to verify AR tracking, MATLAB integration, damage detection, and cockpit inspection functionality.



complete.apk

Future Scope

- Integrate advanced AI models for more accurate aircraft damage detection and predictive maintenance.
- Expand the AR inspection system with cloud connectivity, real-time analytics, and support for multiple aircraft platforms.

Conclusion

The developed AR aircraft inspection system successfully combines Unity, Vuforia, MATLAB image processing, and AI-based analysis for aircraft cockpit maintenance and damage detection. The project improves inspection efficiency by providing real-time AR visualization, damage monitoring, predictive maintenance support, and interactive aircraft component analysis through a mobile application.

THANK YOU!